

* 1 loop correction to Δ_{xy} :

$$\Pi(k^2) = - \left[A + \frac{1}{6} \alpha \left(\frac{1}{\epsilon} + \frac{1}{2} \right) \right] k^2 - \left[B + \alpha \left(\frac{1}{\epsilon} + \frac{1}{2} \right) \right] m^2 + \frac{1}{2} \alpha \int_0^1 dx D \ln(D/\mu^2) + \mathcal{O}(\alpha^2)$$

with $\alpha = g^2/(4\pi)^3$ and $D = x(1-x)k^2 + m^2$

$$\Pi'(k^2) = - \left[A + \frac{1}{6} \alpha \left(\frac{1}{\epsilon} + \frac{1}{2} \right) \right] + \frac{1}{2} \alpha \int_0^1 dx x(1-x) \left(\ln\left(\frac{D}{\mu^2}\right) - 1 \right) + \mathcal{O}(\alpha^2)$$

→ Previously made the requirements that

$$\Pi(-m^2) = 0 \quad \text{and} \quad \Pi'(-m^2) = 0$$

i) First Condition ensures that the exact prop has a pole at $k^2 = -m^2$

ii) The Second ensures that the residue of this pole is one.

→ Normalized as such in order that the LSZ formula hold

Q: But what if there are flaws in the LSZ assumptions which we're missing...

* Consider the massless limit:

$$D = x(1-x)k^2$$

Try to impose: $\Pi(0) = \Pi'(0) = 0$

$$\Pi(0) \rightarrow 0$$

$\Pi'(0)$ is ill defined

$$\lim_{k^2 \rightarrow 0} \left(\ln \left| \frac{x(1-x)k^2}{\mu^2} \right| + 1 \right)$$

• Physically: 1 particle states are no longer separated from the multiparticle continuum by a finite gap in energy

Mathematically: The pole $\tilde{\Delta}(k^2)$ at $k^2 = -m^2$ merges with the branch point at $k^2 = -m^2$ and

is no longer a simple pole

• To remedy this we change the renormalization scheme

→ First go back to $m=0$. Now make a different choice for A and B

$$\boxed{\begin{aligned} \tilde{A} &= -\frac{1}{6}\alpha \frac{1}{\epsilon} + \mathcal{O}(\alpha^2) &= Z_4 - 1 \\ \tilde{B} &= -\alpha \frac{1}{\epsilon} + \mathcal{O}(\alpha^2) &= Z_m - 1 \end{aligned}} \quad (\overline{MS})$$

i.e. we have chosen A and B to cancel the infinities and nothing else.

→ $\tilde{A}$ and $\tilde{B}$ have no finite parts. This is a different normalization scheme.

* Original derivation was the on-shell (OS) scheme.

* The above choice is the Modified Minimal subtraction scheme. $\overline{MS}$

* If we let $\tilde{\mu} = \mu$: Minimal subtraction MS

Now, taking $\Pi(-m^2) = \Pi'(-m^2) = 0$:

$$\Pi_{OS}(k^2) = -\frac{1}{12}\alpha(k^2 + m^2) + \frac{1}{2}\alpha \int_0^1 dx D \ln(D/D_0) + \mathcal{O}(\alpha^2)$$

while

$$\begin{aligned} \Pi_{\overline{MS}}(k^2) &= -\frac{1}{2}\alpha \left[\left(\frac{2}{\epsilon} + 1 \right) \left(\frac{1}{6}k^2 + m^2 \right) + \int_0^1 dx D \ln\left(\frac{M}{D}\right) \right] \\ &\quad - Ak^2 - Bm^2 + \mathcal{O}(\alpha^2) \\ &= \left[-\alpha \left(\frac{1}{\epsilon} + \frac{1}{2} \right) \left(\frac{1}{6} \right) - A \right] k^2 + \left[-\alpha \left(\frac{1}{\epsilon} + \frac{1}{2} \right) - B \right] m^2 + \dots \end{aligned}$$

For the above choices in $\overline{MS}$

$$\begin{aligned} &\left(-\frac{\alpha}{6} \left(\frac{1}{\epsilon} + \frac{1}{2} \right) + \frac{1}{6}\alpha \frac{1}{\epsilon} \right) k^2 + \left[-\alpha \left(\frac{1}{\epsilon} + \frac{1}{2} \right) + \alpha \frac{1}{\epsilon} \right] m^2 \\ &= -\frac{\alpha}{12} k^2 - \frac{\alpha}{2} m^2 - \frac{1}{2}\alpha \int_0^1 dx D \ln(M/D) + \mathcal{O}(\alpha^2) \\ &= -\frac{\alpha}{12} (k^2 + 6m^2) + \frac{1}{2}\alpha \int_0^1 dx D \ln(D/\mu) + \mathcal{O}(\alpha^2) \end{aligned}$$

as claimed in the text

• Note that $\Pi_{\overline{MS}}(k^2)$ has a well defined limit for $m \rightarrow 0$ (i.e. $k^2 \rightarrow 0$) while $\Pi_{OS}(k^2)$ does not. Since $\ln(D/D_0)$ is not defined for $m \rightarrow 0$

$$\lim_{x \rightarrow 0} x \ln(x) = \lim_{x \rightarrow 0} \frac{\ln(x)}{1/x} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{1/x}{-1/x^2} = -\frac{1}{1/x} = x \rightarrow 0$$

• On the other hand $\Pi_{\overline{MS}}$ depends on μ while Π_{OS} does not.

MS: $\Delta_{\overline{MS}}(k^2)$ will no longer have a pole at $k^2 = -m^2$ and instead will be somewhere else.

→ By definition the physical mass m_{ph} is determined by this pole $k^2 = -m_{ph}^2$. Here $m \neq m_{ph}$ and the residue of this pole is no longer 1.

$$\oint \frac{1}{k^2 + m_{ph}^2 + i\epsilon} = 2\pi i R \leftarrow \text{residue}$$

* LSZ formula must now be multiplied by a factor of $R^{-1/2}$ for each external particle

$$\begin{aligned} & \langle 0 | a^\dagger \sqrt{2\omega_p} \varphi(0) | 0 \rangle \\ &= \langle 0 | a^\dagger \sqrt{2\omega_p} \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} (a_k e^{i0} + a_k^\dagger e^{i0}) | 0 \rangle \\ &= \int \frac{d^3k}{(2\pi)^3} (-2\pi i^3 2\omega \delta(k-p)) \langle 0 | 0 \rangle \end{aligned}$$

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I believe this is from $\langle p | \varphi(0) | 0 \rangle \stackrel{!}{=} 1$
 → $R^{-1/2} \varphi(k)$ now has unit amplitude to create a φ state

* Also, in LSZ $-\partial^2 + m^2 \rightarrow -\partial^2 + m_{ph}^2$
 $k_{ext}^2 = -m_{ph}^2 \leftarrow \text{Not parameter in } \mathcal{L}$

Example: Pole in F is a pole in $\mathcal{L}$

Feynman rules. Each external leg now

gives a factor of R

→ Get a net factor of $R^{1/2}$ for each field in $\overline{MS}$.

$$* \quad \tilde{\Delta}_{\overline{MS}}(k) = \frac{1}{k^2 + m^2 - \Pi_{\overline{MS}}(k^2)} \Leftrightarrow \tilde{\Delta}_{\overline{MS}}(k^2)^{-2} = k^2 + m^2 - \Pi_{\overline{MS}}(k^2)$$

$$\Delta_{\overline{MS}}^{-2}(-m_{ph}^2) = 0 \quad (\text{By definition})$$

• Let

$$k^2 = -m_{ph}^2$$

$$m_{ph}^2 = m^2 - \Pi_{\overline{MS}}(-m_{ph}^2)$$

* Since $\Pi_{\overline{MS}}(-m_{ph}^2) \sim \alpha () + \alpha^2 + \dots$

we see that the difference m_{ph}^2 and m^2 is $\mathcal{O}(\alpha)$

or, note this is a recursive relation

$$m_{ph}^2 = m^2 - \Pi_{\overline{MS}}(-m^2 + \mathcal{O}(\alpha))$$

$$= m^2 - \Pi_{\overline{MS}}(-m^2) + \dots$$

$$\left[\begin{aligned} f(x) &= f(a) + (x-a)f'(a) \\ x &= -m^2 + \mathcal{O}(\alpha) \\ a &= -m^2 \\ f(x) &= f(-m^2) + (-m^2 + \mathcal{O}(\alpha) + m^2) f'(-m^2) + \dots \end{aligned} \right.$$

$$= m^2 - (\Pi_{\overline{MS}}(-m^2) + \mathcal{O}(\alpha) \Pi'_{\overline{MS}}(-m^2) + \dots)$$

$$= m^2 - \Pi_{\overline{MS}}(-m^2) + \underbrace{\mathcal{O}(\alpha) \Pi'_{\overline{MS}}}_{=0} + \underbrace{(\mathcal{O}(\alpha))^2 \Pi''_{\overline{MS}}(-m^2)}_{\sim \mathcal{O}(\alpha^2)}$$

$$\text{Since } \Pi'_{\overline{MS}}(-m^2) = 0.$$

$$m_{ph}^2 = m^2 - \Pi_{\overline{MS}}(-m^2) + \mathcal{O}(\alpha^2)$$

$$\Pi_{\overline{MS}}(-m^2) = \frac{\alpha}{12} (-m^2 + 6m^2) + \frac{1}{2} \alpha \int_0^1 dx D_0 \ln\left(\frac{D_0}{\mu}\right) + \mathcal{O}(\alpha^2)$$

$$= \frac{\alpha}{2} \left[\left(\frac{m^2}{\sigma} - m^2 \right) + \int_0^\infty dx D_0 \ln(D_0/\mu) \right] + \mathcal{O}(\alpha^2)$$

Thus,

$$m_{ph}^2 = m^2 - \frac{\alpha}{2} \left[\frac{1}{\sigma} m^2 - m^2 + \int_0^\infty dx D_0 \ln(D_0/\mu) \right] + \mathcal{O}(\alpha^2)$$

$$m_{ph}^2 = m^2 \left[1 + \frac{5}{12} \alpha \left(\ln\left(\frac{\mu^2}{m^2}\right) + c' \right) + \mathcal{O}(\alpha^2) \right]$$

$$c' = (34 - 3\pi\sqrt{3}) = 1.18$$

* Physics should be independent of μ . This means m and α take on different values as μ is varied in such a way that m_{ph} is left invariant

$$\begin{aligned} m &= m(\mu) \\ \alpha &= \alpha(\mu) \end{aligned}$$

$$2 \ln(m_{ph}) = 2 \ln(m) + \ln \left(1 + \frac{5}{12} \alpha \left(\ln\left(\frac{\mu^2}{m^2}\right) + c' \right) + \mathcal{O}(\alpha^2) \right)$$

$$\frac{d}{d \ln(\mu)} \ln(m_{ph}) = \frac{1}{m} \frac{dm}{d \ln(\mu)} + \left(\frac{5}{12} \alpha \right) \frac{d}{d \ln(\mu)} \left(\ln\left(\frac{\mu^2}{m^2}\right) \right)$$

$$\begin{aligned} \ln(m_{ph}) &= 2 \ln(m) \left((\alpha + \alpha^2) + (\alpha + \alpha^2)^2 + \dots \right) \\ &= \ln(m) + (\alpha + \mathcal{O}(\alpha^2)) \end{aligned}$$

$$\frac{d \ln(m_{ph})}{d \ln(\mu)} = 0 = \frac{1}{m} \frac{dm}{d \ln(\mu)} + \left(\frac{5}{12} \alpha - \frac{5}{12} \alpha \frac{1}{m} \frac{dm}{d \ln(\mu)} + \dots \right)$$

assuming $d\alpha/d \ln(\mu) = \mathcal{O}(\alpha^2)$, then

$$\frac{dm}{d \ln(\mu)} = \left(-\frac{5}{12} \alpha + \mathcal{O}(\alpha^2) \right) m$$

we define the anomalous dimension of the mass parameter

$$\gamma_m(\alpha) \equiv -\frac{5}{12} \alpha + \mathcal{O}(\alpha^2)$$

Let $f(z) = N(z)/D(z)$, then : pole at $z = z_0$:

$$c_{-1} = \frac{N(z_0)}{D'(z_0)}$$

$$\hat{z} = \frac{1}{\dots}$$

Here $\Delta(k^2) = k^2 + m^2 - \Pi$

$$\text{Res}(\tilde{\Delta}(k^2)) = R = \frac{1}{\frac{d}{dk^2}(\tilde{\Delta}_{\overline{MS}}(k^2)^{-1})} \Big|_{k^2 = -m_{ph}^2}$$

or

$$R^{-1} = \frac{d}{dk^2}(\tilde{\Delta}_{\overline{MS}}(k^2)^{-1}) \Big|_{k^2 = -m_{ph}^2}$$

$$= 1 - \Pi'_{\overline{MS}}(-m_{ph}^2) = R^{-1}$$

as above we expand using $m_{ph}^2 = m^2 - \Pi'_{\overline{MS}}(-m_{ph}^2) + \dots$

$$R^{-1} = 1 - \Pi'_{\overline{MS}}(-m^2) + \mathcal{O}(\alpha^2)$$

$$\Pi'_{\overline{MS}}(k^2) = -\frac{\alpha}{12} + \frac{1}{2}\alpha \int_0^1 dx (1-x)x \ln(D/\mu^2) + \mathcal{O}(\alpha^2)$$

$$+ \frac{1}{2}\alpha \int_0^1 dx D \frac{1}{D\mu^2} [x(1-x)]$$

$$= -\frac{\alpha}{12} + \frac{1}{2}\alpha \int_0^1 dx x(1-x) \ln(D/\mu^2) + \frac{1}{12}\alpha + \mathcal{O}(\alpha^2)$$

Now,

$$\Pi'_{\overline{MS}}(-m^2) = \frac{1}{2}\alpha \int_0^1 dx x(1-x) \left[\ln\left(\frac{m^2}{\mu^2}\right) + \ln(1-x(1-x)) \right]$$

$$= \frac{1}{12}\alpha \ln(m^2/\mu^2) + \frac{1}{24}(3\sqrt{3}\pi - 17)$$

$$= \frac{1}{12}\alpha \left[\ln(m^2/\mu^2) + \frac{1}{3}(3\sqrt{3}\pi - 17) \right] + \mathcal{O}(\alpha^2)$$

Plugging this back in we have

$$R^{-1} = 1 + \frac{1}{12}\alpha \left[\ln\left(\frac{\mu^2}{m^2}\right) + c'' \right] + \mathcal{O}(\alpha^2)$$

$$c'' = (17 - 3\pi\sqrt{3})/3 \approx 0.23$$

* We can also look at the vertex f_m^u MS:

$$V_3(k_1, k_2, k_3)/g = 1 + \left\{ \alpha \left(\frac{1}{2} + \ln\left(\frac{\mu}{m}\right) + C \right) \right.$$

$$\left. - \frac{1}{2}\alpha \int dF_3 \ln(D/\mu^2) + \mathcal{O}(\alpha^2) \right\}$$

$$\text{MS: Take } C = -\frac{\alpha}{2} + \mathcal{O}(\alpha^2)$$

$$V_{3, \overline{MS}}(k_1, k_2, k_3) = g \left(1 + \alpha \ln\left(\frac{\mu}{m}\right) - \frac{1}{2}\alpha \int dF_3 \ln\left(\frac{D}{\mu^2}\right) + \mathcal{O}(\alpha^2) \right)$$

$$\int dF_3 = 1 \Rightarrow -\frac{1}{2}\alpha \left(\int dF_3 \left(\ln\left(\frac{D}{\mu^2}\right) - \ln\left(\frac{\mu^2}{m^2}\right) \right) \right)$$

$$= -\frac{1}{2}\alpha \int dF_3 \ln(D/\mu^2)$$

$$V_{3, \overline{MS}}(k_1, k_2, k_3) = g \left(1 - \frac{1}{2} \alpha \int dF_3 \ln \left(\frac{\mu^2}{\mu^2} \right) + \mathcal{O}(\alpha^2) \right)$$

$$D = x y k_1^2 + y z k_2^2 + z x k_3^2 + m^2$$

* Now, let's go back and look at $\psi\psi \rightarrow \psi\psi$
 If we replace OS $\rightarrow \overline{MS}$ and add $(R'^{1/2})^4$ external residue factors

$$\tau = R^2 \tau_0 \left[1 - \frac{11}{12} \alpha \left(\ln \left(\frac{s}{\mu^2} \right) + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2) \right]$$

(Go back and calculate τ 's at some pt.)

$$\tau_0 = -g^2 (s^{-1} + t^{-1} + u^{-1})$$

Plugging in $R^{-1} = 1 + \frac{1}{12} \alpha \left(\ln \left(\frac{\mu^2}{m^2} \right) + c'' \right) + \mathcal{O}(\alpha^2)$

$$\tau = \tau_0 \left[1 - \alpha \left(\frac{11}{12} \ln \left(\frac{s}{\mu^2} \right) + \frac{1}{6} \ln \left(\frac{\mu^2}{m^2} \right) + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2) \right]$$

* For a detector with limits bound by δ

$$|\tau_{\text{obs}}|^2 = |\tau|^2 \left[1 + \frac{1}{3} \alpha \left(\ln \left(\frac{\delta^2 s}{m^2} \right) + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2) \right]$$

δ : Angular resolution of the detector

$$|\tau_{\text{obs}}|^2 = |\tau_0|^2 \left[1 - \alpha \left(\frac{3}{2} \ln \left(\frac{s}{\mu^2} \right) + \frac{1}{3} \ln \left(\frac{1}{\delta^2} \right) + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2) \right]$$

* No longer dependent on m !

Well defined for $m \rightarrow 0$

$$\ln |\tau_{\text{obs}}|^2 = \underbrace{C_1 + 2 \ln(\alpha)} + 3 \alpha (\ln(\mu) + C_2) + \mathcal{O}(\alpha^2)$$

• From $\ln |\tau_0|^2 = \ln(\alpha^2 C) = 2 \ln(\alpha) + \ln(C)$

$$\ln(1+\alpha) \approx 3 \alpha \left(\ln(\mu) - \frac{1}{2} \ln(s) - \frac{1}{9} \ln \left(\frac{1}{\delta^2} \right) + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2)$$

$$= 3\alpha (\ln(\mu) + C_2)$$

$$\ln|T_{\text{obs}}|^2 = C_1 + 2\ln(\alpha) + 3\alpha (\ln(\mu) + C_2) + \mathcal{O}(\alpha^2)$$

$$0 = \frac{d}{d\ln(\mu)} \ln|T_{\text{obs}}|^2 = 2 \frac{1}{\alpha} \frac{d\alpha}{d\ln(\mu)} + 3 \frac{d\alpha}{d\ln(\mu)} (\quad)$$

$$+ 3\alpha + \mathcal{O}(\alpha^2)$$

$$* \frac{d\alpha}{d\ln(\mu)} \left(\frac{2}{\alpha} + 3\ln(\mu) + 3C_2 \right) + 3\alpha + \mathcal{O}(\alpha^2) = 0$$

NOT part of solution...

No idea why this isn't part of the solution

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$$0 = \frac{2}{\alpha} \frac{d\alpha}{d\ln(\mu)} + 3\alpha + \mathcal{O}(\alpha^2)$$

$$\frac{d\alpha}{d\ln(\mu)} = -\frac{3}{2}\alpha^2 + \mathcal{O}(\alpha^3) \equiv \beta(\alpha)$$

Beta-function

• choose $\mu^2 \sim s$ (since we have $\ln(\frac{s}{\mu^2})$)

* If we keep only leading terms

$$\frac{d\alpha}{d\ln(\mu)} = -\frac{3}{2}\alpha^2 \Rightarrow -\frac{2}{3} \int_{\alpha(\mu_1)}^{\alpha(\mu_2)} \frac{d\alpha}{\alpha^2} = \int_{\mu_1}^{\mu_2} \ln(\mu)$$

$$\frac{2}{3} \left(\frac{1}{\alpha(\mu_1)} - \frac{1}{\alpha(\mu_2)} \right) = \ln(\mu_2/\mu_1)$$

$$\frac{3}{2} \ln(\mu_2/\mu_1) + \frac{2}{\alpha_1} = \frac{1}{\alpha_2}$$

$$\frac{\frac{3}{2} \ln(\mu_2/\mu_1) \alpha_1 + 1}{\alpha_1} = \frac{1}{\alpha_2}$$

or

$$\alpha(\mu_2) = \frac{\alpha(\mu_1)}{1 + \frac{3}{2} \alpha(\mu_1) \ln(\frac{\mu_2}{\mu_1})}$$

Since $\mu_2 \sim \mu_1$

Suppose $\mu_2 > \mu_1$
 or
 In General: $\alpha(\mu) = 1 + \frac{2}{3} \alpha(\mu_r) \ln\left(\frac{\mu}{\mu_r}\right)$
 as μ increases $\alpha(\mu)$ decreases
 $\alpha(\mu) \xrightarrow{\mu \rightarrow \infty} 0$

↗ If this occurs, then the theory is asymptotically free.

Meaning we can take μ to a value that makes the coupling of the theory small/vanishing / insignificant.

μ^2 : units of mass, i.e. at high energy the theory is trivial and independent of mass ($\mu \rightarrow \infty$)

At lower energies the theory becomes highly coupled

* For $m \neq 0$, process stops at $\mu \sim m$
 Since the maximum value of S is $4m^2$

* $m=0$: $\alpha(\mu)$ continues to increase

• If $\beta(\alpha) = +n\alpha^2 + \mathcal{O}(\alpha^3)$ then the theory is infrared free

Suppose: $\beta(\alpha) = b_1 \alpha^2 + \mathcal{O}(\alpha^3)$
 $\gamma_m(\alpha) = c_1 \alpha + \mathcal{O}(\alpha^2)$

$$\frac{d\alpha}{d \ln(\mu)} = b_1 \alpha^2 \Rightarrow$$

μ_1, μ_2

$$\frac{1}{b_1} \left(\frac{1}{\alpha_2} - \frac{1}{\alpha_1} \right) = - \ln \left(\frac{\mu_2}{\mu_1} \right) \quad \alpha(\mu_1)$$

$$\frac{1}{\alpha_2} = \frac{1 - b_1 \ln(\mu_2/\mu_1)}{\alpha_1} \Rightarrow \boxed{\alpha(\mu_2) = \frac{\alpha(\mu_1)}{1 - b_1 \ln(\mu_2/\mu_1)}}$$

$$\frac{dm}{d \ln(\mu)} = C_1 \alpha m$$

$$d \ln(\mu) = \frac{d \ln(\mu)}{d \alpha} d \alpha$$

$$= \frac{1}{b_1 \alpha^2} d \alpha$$

$$\frac{dm}{dm} = C_1 \alpha m \left(\frac{1}{b_1 \alpha^2} \right) d \alpha$$

$$= \frac{C_1 m}{b_1 \alpha} d \alpha \Rightarrow \int \frac{dm}{m} = \frac{C_1}{b_1} \int \frac{d \alpha}{\alpha}$$

$$\ln \left(\frac{\mu_2}{\mu_1} \right) = \frac{C_1}{b_1} \ln \left(\frac{\alpha_2}{\alpha_1} \right)$$

$$\boxed{m(\mu_2) = \left(\frac{\alpha(\mu_2)}{\alpha(\mu_1)} \right)^{C_1/b_1} m(\mu_1)}$$